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CHEMISTRY

9202/2

Paper 2

Mark scheme

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Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from oxfordaqaexams.org.uk

Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Information to Examiners

1. General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make his or her judgement
- the Assessment Objectives, level of demand and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

2. Emboldening and underlining

2.1 In a list of acceptable answers where more than one mark is available 'any **two** from' is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.

2.2 A bold **and** is used to indicate that both parts of the answer are required to award the mark.

2.3 Alternative answers acceptable for a mark are indicated by the use of **or**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.

2.4 Any wording that is underlined is essential for the marking point to be awarded.

3. Marking points

3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that 'right + wrong = wrong'.

Each error / contradiction negates each correct response. So, if the number of error / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as * in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

Student	Response	Marks awarded
1	green, 5	0
2	red*, 5	1
3	red*, 8	0

Example 2: Name two planets in the solar system.

[2 marks]

Student	Response	Marks awarded
1	Neptune, Mars, Moon	1
2	Neptune, Sun, Mars, Moon	0

3.2 Use of chemical symbols / formulae

If a student writes a chemical symbol / formula instead of a required chemical name, full credit can be given if the symbol / formula is correct and if, in the context of the question, such action is appropriate.

3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. Full marks can, however, be given for a correct numerical answer, without any working shown.

3.4 Interpretation of 'it'

Answers using the word 'it' should be given credit only if it is clear that the 'it' refers to the correct subject.

3.5 Errors carried forward

Any error in the answers to a structured question should be penalised once only.

Papers should be constructed in such a way that the number of times errors can be carried forward is kept to a minimum. Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ecf in the marking scheme.

3.6 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term.

3.7 Brackets

(…) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

3.8 Allow

In the mark scheme additional information, ‘allow’ is used to indicate creditworthy alternative answers.

3.9 Ignore

‘Ignore’ is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

3.10 Do not accept

‘Do **not** accept’ means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

Question 1

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.1	covalent		1	AO1 3.2.1h

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.2	3		1	AO1 3.2.1g

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.3	**C**		1	AO3 3.2.2c 3.2.2e

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.4	distillation		1	AO1 3.10.1.1a 3.10.1.1c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.5	the hydrogen oxidises		1	AO1 3.10.1.2e

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.6	hydrocarbons contain (only) carbon and hydrogen		1	AO3 3.10.1.1b
	(however) ethanol contains oxygen / O	allow (however) ethanol is an alcohol	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.7	$C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$		1	AO2 3.6.1a 3.10.3.1b

Total Question 1		**8**

Question 2

<table>
<tr><th>Question</th><th>Answers</th><th>Extra information</th><th>Mark</th><th>AO/
Spec. Ref.</th></tr>
<tr><td>02.1</td><td><table><tr><th>Name of Particle</th><th>Charge</th><th>Relative mass</th></tr><tr><td>Electron</td><td>–1</td><td>Very small</td></tr><tr><td>Neutron</td><td>0</td><td>1</td></tr><tr><td>Proton</td><td>+1</td><td>1</td></tr></table></td><td>
allow no charge / neutral / zero</td><td>1
1
1</td><td>AO1
3.1.2d
3.1.2j</td></tr>
</table>

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.2	(an atom has) same number of protons and electrons	allow (an atom has) same number of positive and negative charges	1	AO1 3.1.2e

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.3	$^{37}_{17}Cl$		1	AO2 3.1.2f 3.1.2g 3.1.2h

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.4	average mass of the isotopes of the element compared to ^{12}C		1	AO1 3.1.2k

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.5	(mass =) 4 × 20 = 80 (g)		1 1	AO2 3.6.3a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.6	neon atoms have stable arrangement of electrons	allow neon atoms have full shell of electrons allow neon atoms are unreactive	1	AO2 3.1.3c

Total Question 2		**9**

Question 3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.1	(a substance in which atoms) of two or more elements are (chemically) combined		1	AO1 3.2.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.2	alkali metals		1	AO1 3.2.1d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.3	argon		1	AO2 3.2.1b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.4	ions cannot move	allow ions are in a fixed position	1	AO2 3.2.2b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.5	(strong) electrostatic (forces of attraction)	allow (strong) electrostatic bonds	1	AO1 3.2.2a
	between oppositely charged ions	allow between positive and negative ions	1	
	(forces act) in all directions		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.6	$Al(NO_3)_3$		1	AO2 3.2.1c

Total Question 3		**8**

Question 4

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.1	measuring cylinder	allow burette or pipette	1	AO4 3.3.1.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.2	barium (most reactive) magnesium iron copper (least reactive)		1	AO3 3.3.1.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.3	name / type of metal		1	AO4 3.3.1.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.4	metal and (hydrochloric) acid in conical flask measure volume of (hydrogen) gas produced in gas syringe in a fixed time **alternative approach 1** metal and (hydrochloric) acid in conical flask (1) collect a fixed volume of gas in a gas syringe (1) record the time taken (1) **alternative approach 2** stopper / bung **and** conical flask prevent loss of gas (1) stop clock measures (an accurate) time (1) gas syringe measures volume of gas produced (1)		1 1 1	AO4 3.3.1.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.5	barium sulfate is produced (which) is insoluble	 allow (which) is a (white) precipitate	1 1	AO3 3.3.1.1a 3.4.3f

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.6	(white solid) stops the (hydrochloric) acid particles colliding with the metal		1	AO3 3.8.1b

Total Question 4		**9**

Question 5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.1	easier to control temperature with a water bath	allow alkanes are flammable allow Bunsen flame could set alkane on fire	1	AO4 3.10.1.2c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.2	with a smaller hole the cup would empty more slowly	allow converse	1	AO4 3.10.1.2c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.3	$\frac{60 + \mathbf{X} + 55}{3} = 59$		1	AO2 3.10.1.2c
	(**X** =) 62 s		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.4	as the temperature increases the viscosity decreases	allow **1** mark for as the temperature increases the time taken to flow decreases	2	AO3 3.10.1.2c
	change gets smaller as temperature gets higher		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.5	the time taken (to flow out of cup) is larger than expected		1	AO3 3.10.1.2c
	(so therefore) used a larger volume (than 50 cm^3)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.6	(M_r =) $\frac{43.4}{0.31}$		1	AO2 3.6.2c
	= 140		1	
	$\left(\frac{140}{14} =\right)$ 10	allow correct use of an incorrectly determined value of M_r	1	
	$C_{10}H_{20}$		1	
	alternative approach 1: mass of CH_2 = 0.31 x 14 (1) = 4.34 (1) $\left(\frac{43.4}{4.34} =\right)$ 10 (1) $C_{10}H_{20}$ (1)			
	alternative approach 2: moles CH_2 = $\frac{43.4}{14}$ (1) = 3.1 (1) $\left(\frac{3.1}{0.31} =\right)$ 10 (1) $C_{10}H_{20}$ (1)			

Total Question 5		13

Question 6

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.1	sodium moves faster (than lithium) bubbles (more) quickly produced	 allow faster rate of bubbling ignore more bubbles produced	1 1	AO3 3.7.1b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.2	hydrogen	allow H_2	1	AO1 3.7.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.3	purple	allow blue	1	AO2 3.5.1f 3.7.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.4	dissolve solid in water add sodium hydroxide (solution) (iron(II) bromide) green precipitate (iron(III) bromide) brown precipitate	allow solid for precipitate allow red brown precipitate	1 1 1 1	AO4 AO2 3.4.3c

Total Question 6		**8**

Question 7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.1	turns red (then) turns white **or** (then) is bleached		1 1	AO1 3.4.2e

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.2	conducts electricity does not react (with chlorine and / or sodium chloride)		1 1	AO2 AO3 3.3.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.3	hydrogen is less reactive (than sodium)		1	AO1 3.3.2g 3.3.2j

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.4	$2\,H^+ + 2e^- \rightarrow H_2$	allow **1** mark for H^+	2	AO2 3.3.2f

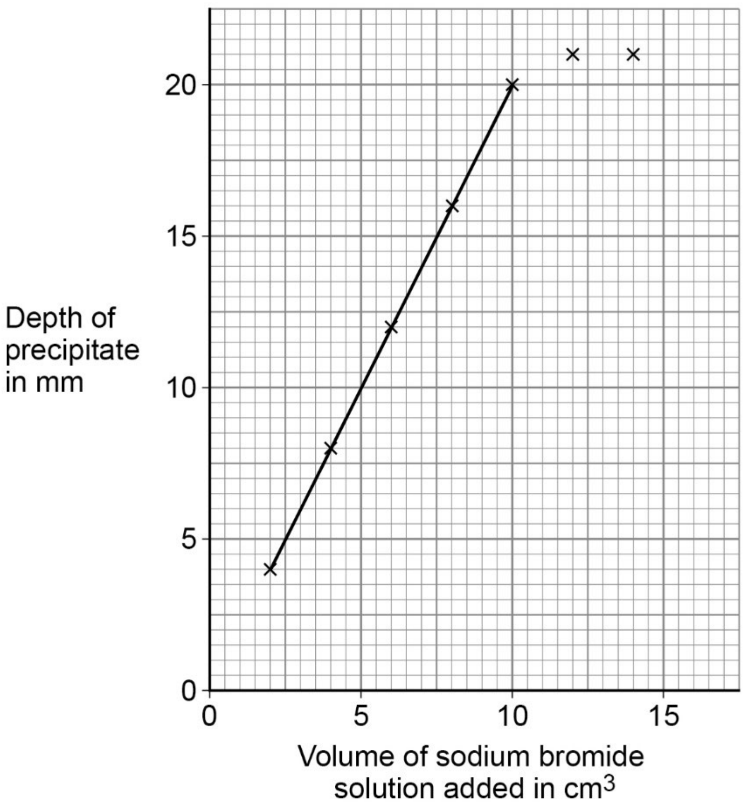
Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.5	NaOH		1	AO2 3.3.2j

Total Question 7		**8**

Question 8

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.1	filter the mixture		1	AO4 3.4.1c 3.5.2c
	wash the residue		1	
	dry the residue (and filter paper)	allow dry the lead bromide	1	
	weigh the (dry) lead bromide	allow weigh the (dry) residue	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.2	to let lead bromide precipitate settle to the bottom of the boiling tube	allow to let the lead bromide precipitate separate (from the mixture)	1	AO4 3.5.2c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.3	scale (5 cm^3 per 2 cm)		1	AO3 × 1 AO2 × 3 3.5.2c
	first 5 points plotted correctly	allow **1** mark for 3 or 4 of first 5 points plotted correctly allow tolerance of ± ½ a small square	2	
	line of best fit		1	
	an answer of  scores **4** marks			

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.4	any **one** from: • all lead nitrate has reacted (with volumes above 10cm^3) • sodium bromide is in excess.		1	AO3 3.5.2c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.5	(conversion 10 cm^3) = 0.010 dm^3		1	AO2 3.6.1d 3.6.4a
	(moles of NaBr =) 0.010 × 0.50	allow correct use of incorrect / no conversion	1	
	= 0.005 (mol)		1	
	(moles of $PbBr_2$ = $\frac{0.005}{2}$ =) 0.0025 (mol)	allow correct use of incorrectly determined moles of NaBr	1	
	(mass of $PbBr_2$ =) 0.0025 × 367	allow correct use of incorrectly determined moles of $PbBr_2$	1	
	=0.9175 (g)	allow 0.92 or 0.918 (g)	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.6	calcium hydroxide solution has a pH greater than 7	allow calcium hydroxide solution is an alkali	1	AO3 3.5.1a 3.5.1f 3.5.2c
	(and) a higher percentage of lead ions are removed (above pH 7)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.7	lead ions are toxic	allow lead ions can make water taste unpleasant allow lead ions can make water discoloured	1	AO3 3.5.2c

Total Question 8		**19**

Question 9

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
09.1	closed system (where the) rate of reaction is the same in each direction		1 1	AO2 3.8.2a

Question	Answers	Mark	AO/ Spec. Ref.
09.2	**Level 3:** Relevant points (reasons / causes) are identified, given in detail and logically linked to form a clear account.	5–6	AO3 × 2 AO2 × 4 3.8.2c 3.8.2d 3.8.2e 3.8.3d
	Level 2: Relevant points (reasons / causes) are identified, and there are attempts at logical linking. The resulting account is not fully clear.	3–4	
	Level 1: Points are identified and stated simply, but their relevance is not clear and there is no attempt at logical linking.	1–2	
	No relevant content	0	
	Indicative content **rate** • higher temperature gives higher rate because of more frequent collisions • higher temperature gives higher rate because more particles have the activation energy • higher pressure gives higher rate because of more frequent collisions • use of catalyst gives higher rate because the activation energy is lowered **equilibrium** • higher temperature decreases the yield because reaction is exothermic • higher pressure increases the yield because more molecules on left-hand side • use of catalyst has no effect on the yield **other factors** • higher temperature (than 450°C) uses more energy so increases costs • higher pressure uses more energy so increases costs • higher pressure requires stronger reaction vessels so increases costs • use of a catalyst reduces energy costs **compromise** • the temperature chosen is a compromise between rate of reaction and position of equilibrium • the temperature chosen is a compromise between rate and cost • the pressure chosen is a compromise between yield / rate and cost • extra cost of higher pressure is not worth extra yield		

Total Question 9		**8**